

Dream Team Analyzer

Project Report

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August 21, 2025

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Abstract

The Dream Team Analyzer is a web-based application that allows football fans to create their dream team, visualize it on a football pitch, and receive AI-generated analysis based on the selected formation. The system uses OpenRouter to interact with Claude 3 Haiku (Anthropic) and provides intelligent suggestions on strengths, weaknesses, and tactical improvements. The frontend is developed using HTML, Tailwind CSS, and JavaScript while Flask serves as the backend.

1. Introduction

Football enthusiasts often enjoy theorizing about dream teams and optimal player placements. This web app allows users to input their preferred lineup, choose a formation, and receive an AI-powered analysis to evaluate its tactical efficiency.

2. Objectives

- To develop an intuitive user interface for team layout on a football field.
- To implement formation-based dynamic input generation.
- To integrate Claude 3 Haiku for tactical team analysis using OpenRouter.
- To allow fast, accessible AI insights for any selected team.

3. Tools and Technologies

- **Frontend:** HTML5, Tailwind CSS, JavaScript
- **Backend:** Python Flask
- **AI API:** Claude 3 Haiku (via OpenRouter)
- **Design Assets:** Football pitch background image

4. System Design

4.1 Architecture Overview

- **Client side:** User enters player names for a formation and submits.
- **Server side:** Flask receives request, formats it, sends to Claude via OpenRouter, and returns the analysis.
- **AI Model:** Claude 3 Haiku processes formation and team lineup, returning strategic insights.

4.2 Data Flow Diagram

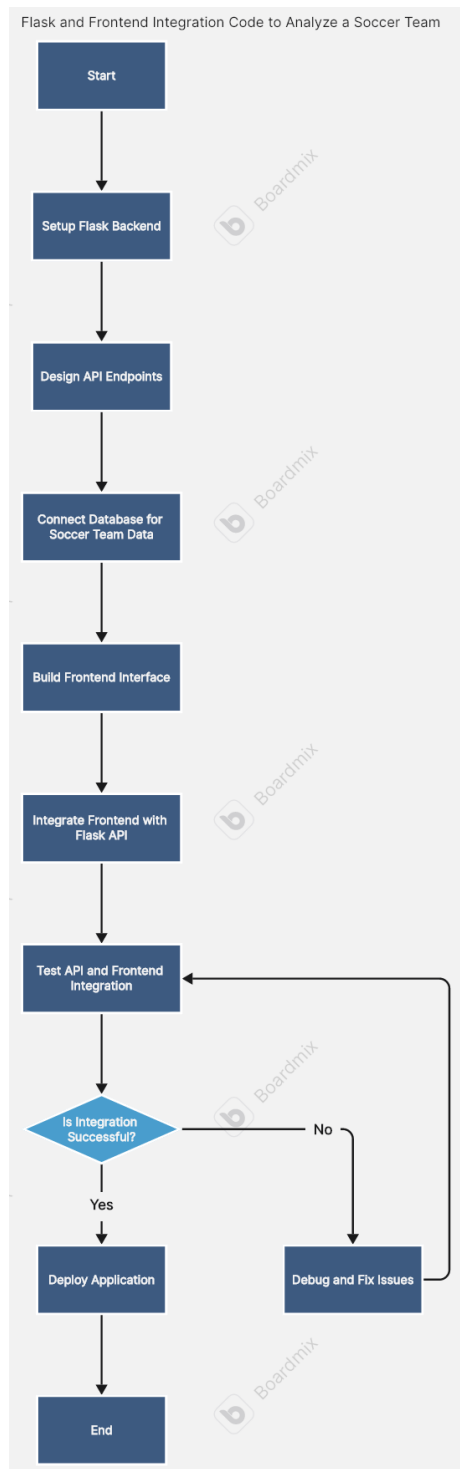


Figure 1: System Data Flow Diagram

5. Implementation

5.1 Frontend Layout

The pitch is displayed using a background image with overlaid input fields positioned based on the formation. JavaScript dynamically updates input positions based on selection.

5.2 Backend Logic

Listing 1: Flask Backend Code

```
1 @app.route("/analyze", methods=["POST"])
2 def analyze_team():
3     data = request.get_json()
4     formation = data.get("formation")
5     players = data.get("team")
6     team_str = "\n".join([f"{pos}: {name}" for pos, name in players.
7                             items()])
8     prompt = f"""You are a football analyst.
9     Analyze this team based on the {formation} formation:
10    {team_str}
11    Give strengths, weaknesses, and suggestions."""
12
13     response = openai.ChatCompletion.create(
14         model="anthropic/claude-3-haiku:beta",
15         messages=[{"role": "user", "content": prompt}],
16         temperature=0.7
17     )
18     return jsonify({"analysis": response["choices"][0]["message"]["content"]})
```

5.3 Frontend Pitch Styling

Listing 2: CSS for pitch background

```
1 .pitch-container {
2     position: relative;
3     width: 100%;
4     max-width: 400px;
5     aspect-ratio: 588 / 1000;
6     background-image: url('/static/img2.jpg');
7     background-size: cover;
8     background-repeat: no-repeat;
9 }
```

6. Working

1. User selects a formation like 4-3-3.
2. Player input fields appear on the image accordingly.
3. After entering names, the user clicks "Analyze Team".
4. Data is sent via POST to Flask API.
5. Flask sends prompt to Claude 3 Haiku.
6. The response is shown in a readable analysis card.

7. Output Screens



Figure 2: Dream Team Analyzer UI



Figure 3: AI-generated team analysis

8. Challenges Faced

- Styling player input fields to align with real pitch zones.
- Handling missing player names before submission.
- Integrating OpenRouter securely with valid headers.
- Debugging CORS and API key restrictions.

9. Future Enhancements

- Integrate player images using football APIs.
- Auto-complete player names from real data.
- Allow drag-and-drop positioning.
- Support saving and sharing team sheets.

10. Conclusion

The Dream Team Analyzer successfully combines frontend interactivity with backend intelligence. It enables real-time football tactical analysis using cutting-edge LLMs, showcasing practical AI use in sports and fan engagement platforms.

11. References

- Project Repository – <https://github.com/Haseeb-zai30/dream-team-analyzer>